

Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. What is output by the following code? (Choose all that apply.)

```
1: public class Fish {
2:     public static void main(String[] args) {
3:         int numFish = 4;
4:         String fishType = "tuna";
5:         String anotherFish = numFish + 1;
6:         System.out.println(anotherFish + " " + fishType);
7:         System.out.println(numFish + " " + 1);
8:     } }
```

- A. 4 1
- B. 5
- C. 5 tuna
- D. 5tuna
- E. 51tuna

F. **The code does not compile.**

Esta es la respuesta correcta porque no puedes asignar un valor numérico a un String.

2. Which of these array declarations are not legal? (Choose all that apply.)

- A. `int[][] scores = new int[5][];`
- B. `Object[][][] cubbies = new Object[3][0][5];`
- C. **`String beans[] = new beans[6];`** beans es el nombre de la variable
- D. `java.util.Date[] dates[] = new java.util.Date[2][];`
- E. **`int[][] types = new int[];`** No indica el tamaño de la primera dimensión
- F. **`int[][] java = new int[][];`** No indica el tamaño de la primera dimensión

3. Note that March 13, 2022 is the weekend when we spring forward, and November 6, 2022 is when we fall back for daylight saving time. Which of the following can fill in the **blank** without the code throwing an exception? (Choose all that apply.)

```
var zone = ZoneId.of("US/Eastern");
var date = _____;
var time = LocalTime.of(2, 15);
var z = ZonedDateTime.of(date, time, zone);
```

- A. **`LocalDate.of(2022, 3, 13)`**
- B. `LocalDate.of(2022, 3, 40)`
- C. **`LocalDate.of(2022, 11, 6)`**

La fecha existe. Aunque las 2:15 no existen por el cambio al horario de verano, Java ajusta la hora automáticamente.

La fecha existe. Durante el cambio de otoño, esa hora sigue siendo válida.

D. `LocalDate.of(2022, 11, 7)`

Es una fecha y hora normales, sin problema de horario.

E. `LocalDate.of(2023, 2, 29)`

F. `LocalDate.of(2022, MonthEnum.MARCH, 13);`

4. Which of the following are output by this code? (Choose all that apply.)

```
3: var s = "Hello";
4: var t = new String(s);
5: if ("Hello".equals(s)) System.out.println("one");
6: if (t == s) System.out.println("two");
7: if (t.intern() == s) System.out.println("three");
8: if ("Hello" == s) System.out.println("four");
9: if ("Hello".intern() == t) System.out.println("five");
```

A. `one`

"Hello".equals(s) compara el contenido. Ambos contienen "Hello", por eso imprime one

B. `two`

C. `three`

C: t.intern() devuelve la referencia de "Hello" guardada en el String Pool, que es la misma referencia que usa s. Por eso imprime three.

D. `four`

Tanto el literal "Hello" como s apuntan al mismo objeto del String Pool. Por eso == da true e imprime four.

E. `five`

F. The code does not compile.

G. None of the above

5. What is the result of the following code?

```
7: var sb = new StringBuilder();
8: sb.append("aaa").insert(1, "bb").insert(4, "ccc");
9: System.out.println(sb);
```

A. `abbaaccc`

B. `abbaccca`

Primero queda aaa; luego inserta bb en el índice 1-> abbaa; finalmente inserta ccc en el índice 4-> abbaaccca

C. `bbaaaccc`

D. `bbaaccca`

E. An empty line

F. The code does not compile.

6. How many of these lines contain a compiler error? (Choose all that apply.)

```
23: double one = Math.pow(1, 2);
24: int two = Math.round(1.0);
25: float three = Math.random();
26: var doubles = new double[] {one, two, three};
```

A. 0

B. 1

- C. **2** `Math.round(1.0)` recibe un `double`, por lo que devuelve `long`; no puede asignarse directamente a un `int`.
`Math.random()` devuelve `double`; no puede asignarse directamente a `float`.
- D. 3
- E. 4

7. Which of these statements is true of the two values? (Choose all that apply.)

2022-08-28T05:00 GMT-04:00

2022-08-28T09:00 GMT-06:00

- A. **The first date/time is earlier.**
- B. The second date/time is earlier.
- C. Both date/times are the same.
- D. The date/times are two hours apart.
- E. **The date/times are six hours apart.**
- F. The date/times are 10 hours apart.

8. Which of the following return 5 when run independently? (Choose all that apply.)

```
var string = "12345";
```

```
var builder = new StringBuilder("12345");
```

- A. **builder.charAt(4)** `charAt(4)` busca el carácter en el índice 4. Como los índices empiezan en 0, devuelve '5'.
- B. **builder.replace(2, 4, "6").charAt(3)** `replace(2, 4, "6")` cambia los índices 2 y 3. "12345" queda "1265"; después `charAt(3)` devuelve '5'.
- C. `builder.replace(2, 5, "6").charAt(2)`
- D. `string.charAt(5)`
- E. `string.length`
- F. **string.replace("123", "1").charAt(2)** `replace("123", "1")` convierte "12345" en "145"; después `charAt(2)` devuelve '5'.
- G. None of the above

9. Which of the following are true about arrays? (Choose all that apply.)

- A. **The first element is index 0.**
- B. The first element is index 1.
- C. **Arrays are fixed size.**
- D. Arrays are immutable.
- E. Calling `equals()` on two different arrays containing the same primitive values always returns `true`.
- F. **Calling `equals()` on two different arrays containing the same primitive values always returns `false`.**
- G. Calling `equals()` on two different arrays containing the same primitive values can return `true` or `false`.

10. How many of these lines contain a compiler error? (Choose all that apply.)

```
23: int one = Math.min(5, 3);  
24: long two = Math.round(5.5);  
25: double three = Math.floor(6.6);  
26: var doubles = new double[] {one, two, three};
```

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

11. What is the output of the following code?

```
var date = LocalDate.of(2022, 4, 3);  
date.plusDays(2);  
date.plusHours(3);  
System.out.println(date.getYear() + " " + date.getMonth()  
    + " " + date.getDayOfMonth());
```

- A. 2022 MARCH 4
- B. 2022 MARCH 6
- C. 2022 APRIL 3
- D. 2022 APRIL 5
- E. The code does not compile. LocalDate solo maneja fecha, no hora. Por eso no existe plusHours(3) para ese tipo de objeto.
- F. A runtime exception is thrown.

12. What is output by the following code? (Choose all that apply.)

```
var numbers = "012345678".indent(1);  
numbers = numbers.stripLeading();  
System.out.println(numbers.substring(1, 3));  
System.out.println(numbers.substring(7, 7));  
System.out.print(numbers.substring(7));
```

- A. 12 substring(1, 3) toma desde el índice 1 hasta antes del 3, por eso devuelve "12".
- B. 123
- C. 7
- D. 78 substring(7) toma desde el índice 7 hasta el final, por eso devuelve "78".
- E. A blank line substring(7, 7) devuelve una cadena vacía, así que imprime una línea en blanco.
- F. The code does not compile.
- G. An exception is thrown.

13. What is the result of the following code?

```
public class Lion {
    public void roar(String roar1, StringBuilder roar2) {
        roar1.concat("!!!");
        roar2.append("!!!");
    }
    public static void main(String[] args) {
        var roar1 = "roar";
        var roar2 = new StringBuilder("roar");
        new Lion().roar(roar1, roar2);
        System.out.println(roar1 + " " + roar2);
    } }
```

- A. roar roar
- B. **roar roar!!!** String es inmutable, por eso roar1 no cambia. StringBuilder sí es mutable y append() agrega !!!.
- C. roar!!! roar
- D. roar!!! roar!!!
- E. An exception is thrown.
- F. The code does not compile.

14. Given the following, which can correctly fill in the blank? (Choose all that apply.)

```
var date = LocalDate.now();
var time = LocalTime.now();
var dateTime = LocalDateTime.now();
var zoneId = ZoneId.systemDefault();
var zonedDateTime = ZonedDateTime.of(dateTime, zoneId);
Instant instant = _____;
```

- A. **Instant.now()** Instant.now() crea correctamente el instante actual.
- B. new Instant()
- C. date.toInstant()
- D. dateTime.toInstant()
- E. time.toInstant()
- F. **zonedDateTime.toInstant()** zonedDateTime.toInstant() funciona porque el objeto ya contiene fecha, hora y zona horaria.

15. What is the output of the following? (Choose all that apply.)

```
var arr = new String[] { "PIG", "pig", "123"};
Arrays.sort(arr);
System.out.println(Arrays.toString(arr));
System.out.println(Arrays.binarySearch(arr, "Pippa"));
```

A. [pig, PIG, 123]

B. [PIG, pig, 123]

C. [123, PIG, pig]

Al ordenar, los números quedan antes que las letras y las mayúsculas antes que las minúsculas: [123, PIG, pig].

D. [123, pig, PIG]

E. -3

F. -2

"Pippa" debería insertarse en el índice 1, así que `binarySearch()` devuelve -2.

G. The results of `binarySearch()` are undefined in this example.

16. What is included in the output of the following code? (Choose all that apply.)

```
var base = "ewe\nsheep\\t";
int length = base.length();
int indent = base.indent(2).length();
int translate = base.translateEscapes().length();
```

```
var formatted = "%s %s %s".formatted(length, indent, translate);
System.out.format(formatted);
```

A. 10

`translateEscapes()` convierte `\t` en una tabulación real, reduciendo la longitud a 10.

B. 11

La cadena original tiene 11 caracteres.

C. 12

D. 13

E. 14

F. 15

G. 16

`indent(2)` agrega espacios a cada línea y un salto final, dejando longitud 16.

17. Which of these statements are true? (Choose all that apply.)

```
var letters = new StringBuilder("abcdefg");
```

A. `letters.substring(1, 2)` returns a single-character `String`.

B. `letters.substring(2, 2)` returns a single-character `String`.

C. `letters.substring(6, 5)` returns a single-character `String`.

D. `letters.substring(6, 6)` returns a single-character `String`.

E. `letters.substring(1, 2)` throws an exception.

F. `letters.substring(2, 2)` throws an exception.

G. `letters.substring(6, 5)` throws an exception.

H. `letters.substring(6, 6)` throws an exception.

18. What is the result of the following code? (Choose all that apply.)

```
13: String s1 = ""
```

```
14:   purr"";
```

```

15: String s2 = "";
16:
17: s1.toUpperCase();
18: s1.trim();
19: s1.substring(1, 3);
20: s1 += "two";
21:
22: s2 += 2;
23: s2 += 'c';
24: s2 += false;
25:
26: if ( s2 == "2cfalse") System.out.println("==");
27: if ( s2.equals("2cfalse")) System.out.println("equals");
28: System.out.println(s1.length());

```

- A. 2
- B. 4
- C. **7** s1 termina como "purrtwo", que tiene 7 caracteres.
- D. 10
- E. ==
- F. **equals** equals() compara contenido, y s2 contiene "2cfalse".
- G. An exception is thrown.
- H. The code does not compile.

19. Which of the following fill in the blank to print a positive integer? (Choose all that apply.)

```

String[] s1 = { "Camel", "Peacock", "Llama"};
String[] s2 = { "Camel", "Llama", "Peacock"};
String[] s3 = { "Camel"};
String[] s4 = { "Camel", null};
System.out.println(Arrays._____);

```

- A. **compare(s1, s2)** compare(s1, s2) devuelve positivo porque "Peacock" es mayor que "Llama" en la primera diferencia.
- B. **mismatch(s1, s2)** mismatch(s1, s2) devuelve 1 porque difieren por primera vez en el índice 1.
- C. compare(s3, s4)
- D. **mismatch (s3, s4)** mismatch(s3, s4) también devuelve 1 porque ahí aparece la primera diferencia.
- E. compare(s4, s4)
- F. mismatch (s4, s4)

- 20.** Note that March 13, 2022 is the weekend that clocks spring ahead for daylight saving time. What is the output of the following? (Choose all that apply.)

```
var date = LocalDate.of(2022, Month.MARCH, 13);
var time = LocalTime.of(1, 30);
var zone = ZoneId.of("US/Eastern");
var dateTime1 = ZonedDateTime.of(date, time, zone);
var dateTime2 = dateTime1.plus(1, ChronoUnit.HOURS);

long diff = ChronoUnit.HOURS.between(dateTime1, dateTime2);
int hour = dateTime2.getHour();
boolean offset = dateTime1.getOffset()
    == dateTime2.getOffset();
System.out.println("diff = " + diff);
System.out.println("hour = " + hour);
System.out.println("offset = " + offset);
```

- A. `diff = 1` Entre ambos objetos transcurre una hora real.
- B. `diff = 2`
- C. `hour = 2`
- D. `hour = 3` Al sumar una hora, Java salta de 1:30 a 3:30 por el cambio de horario.
- E. `offset = true`
- F. The code does not compile.
- G. A runtime exception is thrown.

- 21.** Which of the following can fill in the blank to print `avaJ`? (Choose all that apply.)

```
3: var puzzle = new StringBuilder("Java");
4: puzzle._____;
5: System.out.println(puzzle);
```

- A. `reverse()` `reverse()` convierte "Java" directamente en "avaJ".
- B. `append("vaJ$").substring(0, 4)`
- C. `append("vaJ$").delete(0, 3).deleteCharAt(puzzle.length() - 1)`
- D. `append("vaJ$").delete(0, 3).deleteCharAt(puzzle.length())`
- E. None of the above

La cadena se transforma paso a paso hasta quedar en "avaJ".

- 22.** What is the output of the following code?

```
var date = LocalDate.of(2022, Month.APRIL, 30);
date.plusDays(2);
date.plusYears(3);
System.out.println(date.getYear() + " " + date.getMonth()
    + " " + date.getDayOfMonth());
```


- A. `2022 APRIL 30` LocalDate es immutable y los resultados de `plusDays()` y `plusYears()` no se guardan, así que la fecha permanece igual.
- B. `2022 MAY 2`
- C. `2025 APRIL 2`
- D. `2025 APRIL 30`
- E. `2025 MAY 2`
- F. The code does not compile.
- G. A runtime exception is thrown.